

What is a Database?

A database is an organized collection of data that can be easily accessed, managed, and updated. It is used to store, retrieve and manipulate data efficiently and securely.

Importance of Databases:

- Help store large volumes of structured information.
- Allows fast searching, sorting, and filtering of data.
- Maintains data accuracy and consistency.
- Enables multiple users to access data simultaneously.
- It plays a critical role in modern software systems.

Types of Databases:

1. Relational Databases (RDBMS): Use tables with rows and columns (e.g., MySQL, PostgreSQL).
2. NoSQL Databases: Use documents or key-value pairs
3. Cloud Databases: Hosted on cloud platforms and accessible via the internet (e.g., Firebase).

What is Normalization?

Normalization is a process in database design that organizes columns and tables to reduce data redundancy and improve data integrity. It divides large complex tables into smaller, simpler ones and defines clear relationships between them.

First Normal Form (1NF)

- Each column contains atomic (indivisible) values.
- Each record (row) is unique.
- There are no repeating groups of data.

Example: Do not store multiple phone numbers in one column. Use separate rows or columns.

Second Normal Form (2NF)

- Must satisfy 1NF.
- Every non-key column must depend on the entire primary key (not just part of it).
- Eliminates partial dependencies in tables with composite keys.

Example: In a table with a composite key (StudentID, SubjectID), the grade should depend on both keys, not just one.

Third Normal Form (3NF)

- Must satisfy 2NF.
- No transitive dependencies: A non-key column should not depend on another non-key column.
- Ensures that all data in a table relates directly to the primary key.

Example: If a student table has a “ClassName” and “ClassID”, and “ClassName” can be derived from “ClassID”, move it to a separate Class table.